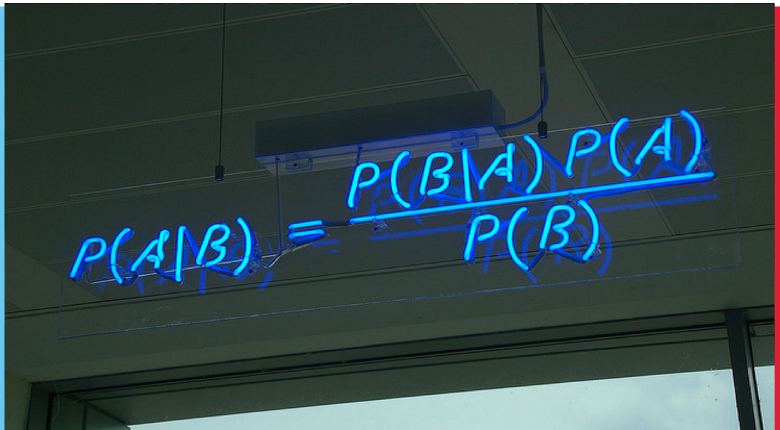
Computer Science Department

Applied Machine Learning

Week3: Bayesian Classification



Dr Haithem Afli

@AfliHaithem

Haithem.afli@mtu.ie

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Contents

Conditional Probability

- Unconditional probability deals with the independent probability of a proposition. That is the probability of a proposition that is not contingent on any other probability.

- Conditional Probability:
  - Represents the probability that one **event will occur given that a second event has already occurred**

```
P(cavity | toothache) = 0.8
```

"Prob. of cavity is 0.8, given that all you know is you have toothache"

Conditional Probability

$$P(B|A)$$

Event that has already occurred

Event that may or may not occur

The line indicates “Given"

Product Rule

- The product rule can be applied

$$P(a \wedge b) = P(a \mid b)\, P(b) = P(b \mid a)\, P(a)$$

- What is the probability of drawing two kings from a deck of card (note the first king is not replaced before picking the second)

- a = first card is a king
- b = second card is a king
- P(a and b) = P(b|a) P(a)

Bayes' Rule

$$P(c \mid d) = (P(d \mid c) \, P(c)) \, / \, P(d)$$

Bayes' Rule can be easily derived from the product rule

$P(c \wedge d) = \mathbf{P(c \mid d)\, P(d) = P(d \mid c)\, P(c)}$

Divide by P(d): $\mathbf{P(c \mid d) = P(d \mid c)\, P(c)/P(d)}$

Let’s Apply it

Example

P(c | d) = (P(d | c) P(c)) / P(d)

- A prison has a population of 600 inmates. It has two cellblocks.
- Cell block A has 500 prisoners with the remainder of the prisoners in cell block B.
- Half the prisoners in cell black A wear a red uniform. The other half wears a blue uniform.
- All prisoners in cell block B wear a blue uniform.
- A prisoner wearing a blue uniform is seen escaping. What is the probability that the prisoner is from cell block A?

P(CellBlockA | BlueU)

=

Example

```
P(c | d) = ( P(d | c) P(c) ) / P(d)
```

- A prison has a population of 600 inmates. It has two cellblocks.
- Cell block A has 500 prisoners with the remainder of the prisoners in cell block B.
- Half the prisoners in cell black A wear a red uniform. The other half wears a blue uniform.
- All prisoners in cell block B wear a blue uniform.
- A prisoner wearing a blue uniform is seen escaping. What is the probability that the prisoner is from cell block A?

P(CellBlockA | BlueU)

= P(BlueU | CellBlockA)*P(CellBlockA)/P(BlueU)

= (((250/500)(500/600)) / (350/600))
= 0.71

Contents

Classification Example

- The table below shows a database of names for workers that work in a particular company. If I pick an employee from the company and their name is Joe, is this individual male or female? The class in this problem is the Sex and the attribute/feature is the Name of the individual.
- This is a trivial problem but we can use Bayes to help solve it.

Name	Sex
Joe	**Female**
Jim	Male
Joe	**Male**
Ted	Male
Mary	Female
Joe	**Male**
Carol	Female
Emily	Female

Classification Example

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Name	Sex
Joe	Female
Jim	Male
Joe	Male
Ted	Male
Mary	Female
Joe	Male
Carol	Female
Emily	Female

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

- We must calculate the probability for each class and then adopt the class with the highest probability
- In other words calculate probability of being a male/female given that individual is called Joe.
  - P(Female | Joe)
  - P(Male | Joe)

Classification Example

Name	Sex
Joe	Female
Jim	Male
Joe	Male
Ted	Male
Mary	Female
Joe	Male
Carol	Female
Emily	Female

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

- **P(Female | Joe)** = P(Joe | Female)P(Female) / P(Joe)
  = (1/4)(4/8) / (3/8) = **0.333**
- **P(Male | Joe)** = P(Joe | Male)P(Male) / P(Joe)
  = (2/4)(4/8)/(3/8) = **0.666**
- Probability of Joe being a Male is higher, therefore we can classify the individual Joe as being a Male

Classification Example

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

- P(Female | Joe) = P(Joe | Female)P(Female) / P(Joe)
  = (1/4)(4/8) / **(3/8)** = 0.333

- P(Male | Joe) = P(Joe | Male)P(Male) / P(Joe)
  = (2/4)(4/8) / **(3/8)** = 0.666

You might notice that for all these calculations, the denominators are identical—P(d). Thus, they are independent of the hypotheses.

Classification Example (maximisation)

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

- P(Female | Joe) = P(Joe | Female)P(Female) / P(Joe)

  = (1/4)(4/8) = 0.125

- P(Male | Joe) = P(Joe | Male)P(Male) / P(Joe)

  = (2/4)(4/8) = 0.25

More Formally

- More formally we want to identify the most likely class C_{MAP}
- For all classes the one that maximises probability of that class given the attribute d

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(c \mid d)$$

$$= \underset{c \in C}{\operatorname{argmax}} \frac{P(d \mid c) P(c)}{P(d)}$$

$$= \underset{c \in C}{\operatorname{argmax}} P(d \mid c) P(c)$$

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Bayes with Multiple Attributes/Features

Bayes with Multiple Attributes/Features

- In the previous slides we considered Bayes classification when we had only a single feature (for example the name of an individual).

- But what if we have many other attributes as well such as age, height, weight etc.

Height	Weight	Name	Sex
X	X	Joe	Female
X	X	Jim	Male
X	X	Joe	Male
X	X	Ted	Male
X	X	Mary	Female
X	X	Joe	Male
X	X	Carol	Female
x	X	Emily	Female

Bayes with Multiple Attributes/Features

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(d \mid c) P(c)$$

$$= \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

Height	Weight	Name	Sex
X	X	Joe	Female
X	X	Jim	Male
X	X	Joe	Male
X	X	Ted	Male
X	X	Mary	Female
X	X	Joe	Male
X	X	Carol	Female
x	X	Emily	Female

- We denote the n features
  X1, X2, .., Xn

Naïve Bayes

$$P(x_1, x_2, \ldots, x_n \mid c)$$

- We make the naïve assumption of conditional independence
  - We assume the feature probabilities $P(x_i \mid c_j)$ are independent given a class c.
  - Whether one features occurs given a class and whether another feature occurs given a class are independent

$$P(x_1, \ldots, x_n \mid c) = P(x_1 \mid c) \bullet P(x_2 \mid c) \bullet P(x_3 \mid c) \bullet \ldots \bullet P(x_n \mid c)$$

Naïve Bayes

- Therefore, we can reformulate our Bayesian classification equation as:

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c) \prod_{x \in X} P(x \mid c)$$

- Let's return to the previous example (with the multi-attribute dataset).
- Assume we pick an employee with the following attribute values
- **Name= "Joe", Weight = 13.2, Height = 6.1**.
  - Should we class this individual as a male or female? How would we work this out?

Succeeding Together

Assume we pick an employee with the following attribute values **Name= "Joe", Weight = 13.2, Height = 6.1**.

Should we class this individual as a male or female? How would we solve this?

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c) \prod_{x \in X} P(x \mid c)$$

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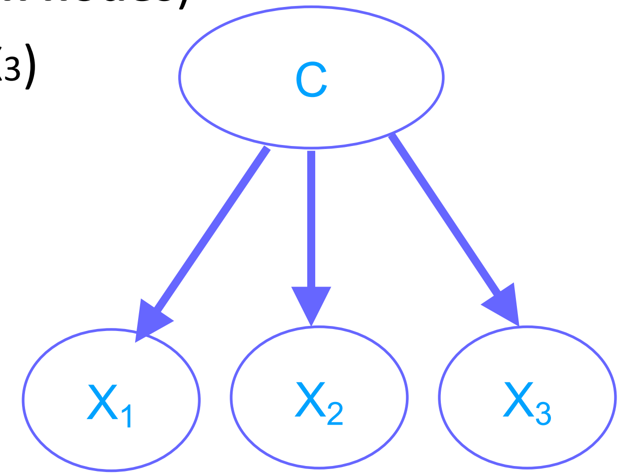
Height	Weight	Name	Sex
X	X	Joe	Female
X	X	Jim	Male
X	X	Joe	Male
X	X	Ted	Male
X	X	Mary	Female
X	X	Joe	Male
X	X	Carol	Female
x	X	Emily	Female

P(Female | Name= "Joe", Weight = 13.2, Height = 6.1) = P(Name = "Joe" | Female) * P(**Weight = 13.2** | Female) * P(Height = 6.1 | Female) * P(Female)

P(Male | Name= "Joe", Weight = 13.2, Height = 6.1) = P(Name = "Joe" | Male) * P(**Weight = 13.2** | Male) * P(Height = 6.1 | Male) * P(Male)

Naïve Bayes Model

- Simplest form of Bayesian classifier (assume **boolean nodes**)
  - A node for each feature in the domain (X_1, X_2, X_3)
  - C is a Boolean class node



- For each arc between the class variable and an evidence node we need a set of four probabilities. These are:
  - P(X1=true | C=true)
  - P(X1=true | C=false)
  - P(X1=false | C=true)
  - P(X1=false | C=false)

Training a Naïve Bayes Classifier

- We estimate conditional probabilities for each arc from the training data!

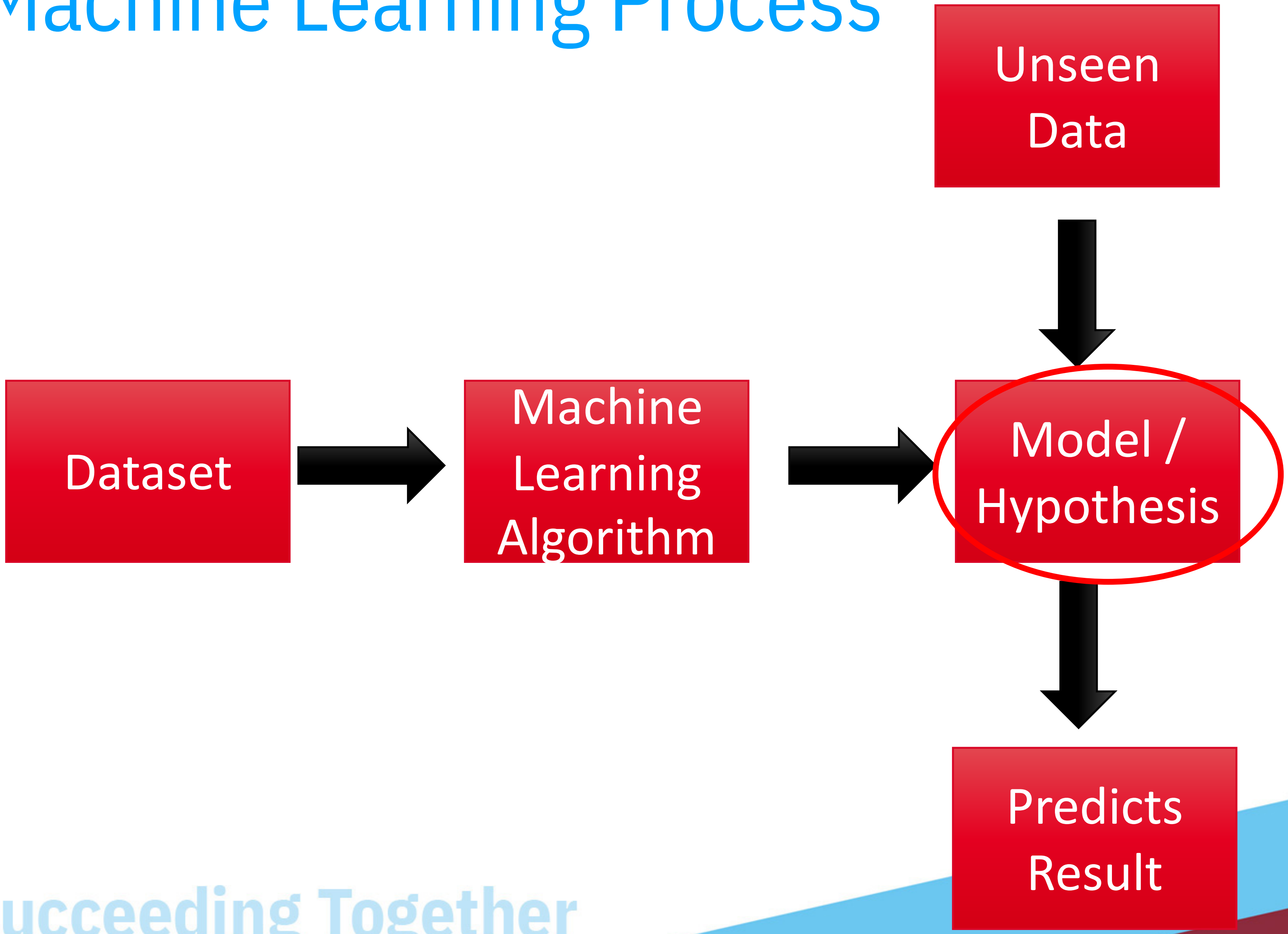
- Simple frequency estimates:
  - $\mathbf{P(X=x_1 \mid C=c_1) = N_{x1c1} / N_{c1}}$
  - N_{x1c1} = counts of cases where $X=x_1$ and $C=c_1$
  - N_{c1} = count of cases where $C=c_1$

P(name = “Joe” | class = male) =

How many of male class are called Joe / Total number male class

Discussion

Machine Learning Process



Classification

Let us understand Naive Bayes with the help of an example

All possible weather combinations

- Summer | Monsoon | Winter
- Sunny | No Sun
- Windy | No Wind

Naïve Bayes Classifier Example (Weather Dataset)

In order to see the probability estimates in action we will look at the weather dataset. We will need to answer questions such as the following. Given the information I know, will I go playing Tennis?

Outlook =sunny, Temp = cold, Humidity = high, Windy = true: **Play = ?**

Anyone for Tennis?

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Outlook =sunny, Temp = cold, Humidity = high, Windy = true: **Play = ?**

Naïve Bayes Classifier Example (Weather Dataset)

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In order to see the probability estimates in action we will look at the weather dataset.

Outlook =<u>sunny</u>, Temp = <u>cold</u>, Humidity = <u>high</u>, Windy = <u>true</u>: Play = ?

To answer this question we will first need to work out a whole set of probabilities.

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c) \prod_{x \in X} P(x \mid c)$$

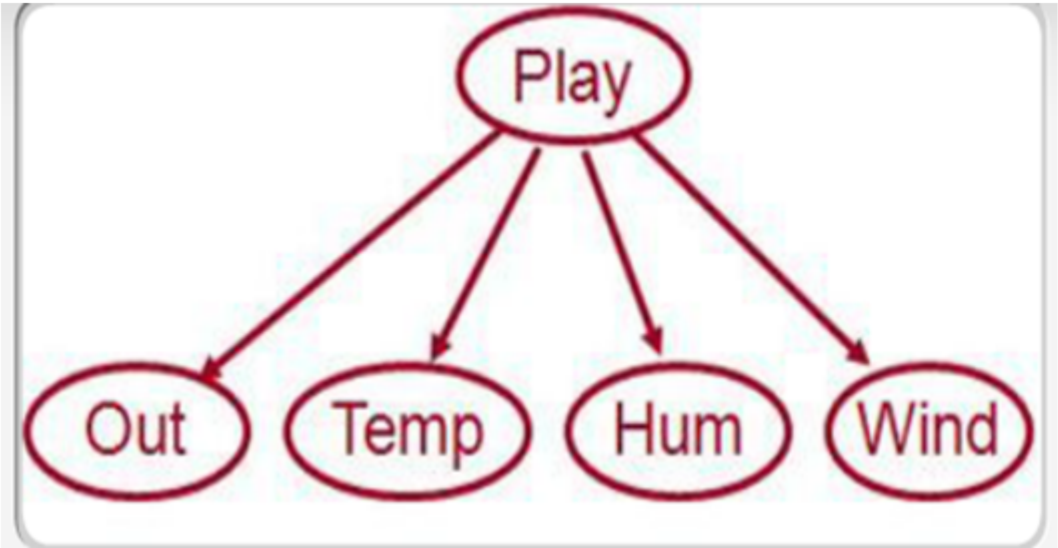
P(Play = y **| Outlook** = sunny, **Temp** = cold, **Humidity** = high, **Windy** = true**)** =

P(Play = y) * P(Outlook=s | Play=y)*P(Temp=c | Play=y)*P(Humidity = h | Play=y)*P(Windy=t | Play=y)

P(Play = n **| Outlook** = sunny, **Temp** = cold, **Humidity** = high, **Windy** = true**)** =

P(Play = n) * P(Outlook=s | Play=n)*P(Temp=c | Play=n)*P(Humidity = h | Play=n)*P(Windy=t | Play=n)

Naïve Bayes Classifier Example



P(Out = Sunny | Play = Y)

P(Out = Overcast | Play = Y)

P(Out = Rainy | Play = Y)

P(Temp = Hot | Play = Y)

P(Temp = Mild | Play = Y)

P(Temp = Cool | Play = Y)

P(Hum= High | Play = Y)

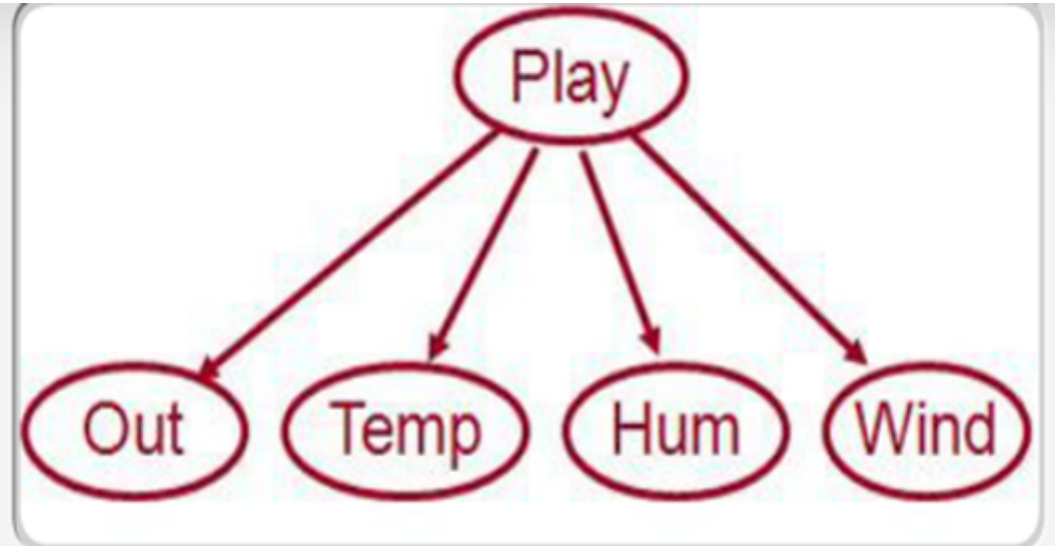
P(Hum= Normal | Play = Y)

P(Win= False | Play = Y)

P(Win= True | Play = Y)

Naïve Bayes Classifier Example

Conditional Probabilities also need to be worked out for Play = N



P(Out = Sunny | Play = N)

P(Out = Overcast | Play = N)

P(Out = Rainy | Play = N)

P(Temp = Hot | Play = N)

P(Temp = Mild | Play = N)

P(Temp = Cool | Play = N)

P(Hum= High | Play = N)

P(Hum= Normal | Play = N)

P(Win= False | Play = N)

P(Win= True | Play = N)

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

- We will first look at calculating the probabilities needed for the class 'play' and the attribute '**windy**'. Lets work out the P(Wind = t | Play = y)
  - $P(X=x_1 \mid C=c_1) = N_{x1c1} / N_{c1}$
  - N_{x1c1} = counts of cases where $X=x_1$ and $C=c_1$
  - N_{c1} = count of cases where $C=c_1$

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

- We will first look at calculating the probabilities needed for the class 'play' and the attribute '**windy**'. Lets work out the P(Wind = t | Play = y)

As we can see from the image, there are 9 cases where `Play=y`. In 3 of these, `Wind=t`, and in the other 6, `Wind=f`. Therefore, the probability of `Wind=t` given that `Play=y` is 3/9, according to these observations.

```
P(Wind=t | Play=y) = 3/9
P(Wind=f | Play=y) = 6/9
```

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

Next work out P(Wind = t | Play = n) and P(Wind = f | Play = n)

As we can see from the image, there are 5 cases where `Play=n`. In 3 of these, `Wind=t`, and in the other 2, `Wind=f`. Therefore:

```
P(Wind=t | Play=n) = 3/5
P(Wind=f | Play=n) = 2/5
```

We now have all four probabilities we need for the arc between play and windy. Next we need to apply the same method to calculate the probabilities for each of the other arcs.

ID	Outlook	Temp	Humidity	Windy	Play?
A	sunny	hot	high	false	no
B	sunny	hot	high	true	no
C	overcast	hot	high	false	yes
D	rainy	mild	high	false	yes
E	rainy	cool	normal	false	yes
F	rainy	cool	normal	true	no
G	overcast	cool	normal	true	yes
H	sunny	mild	high	false	no
I	sunny	cool	normal	false	yes
J	rainy	mild	normal	false	yes
K	sunny	mild	normal	true	yes
L	overcast	mild	high	true	yes
M	overcast	hot	normal	false	yes
N	rainy	mild	high	true	no

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Calculate probabilities for Humidity Attribute
Start with:
P(Humidity =high| Play = yes

P(Humidity =high| Play = yes) = 3/9
P(Humidity =normal| Play = yes) = 6/9

P(Humidity =high| Play = no) = 4/5
P(Humidity =normal| Play = no) = 1/5

Naïve Bayes Example

Using the same approach we can calculate the probabilities associated with each of the other arcs.

<u>Probabilities associated with play and evidence arc Outlook.</u>

```
P(Outlook=s | Play=y) = 2/9  P(Outlook=s | Play=n) = 3/5
P(Outlook=o | Play=y) = 4/9  P(Outlook=o | Play=n) = 0/5
P(Outlook=r  | Play=y) = 3/9 P(Outlook=r  | Play=n) = 2/5
```

<u>Probabilities associated with play and evidence arc Temp.</u>

```
P(Temp=h | Play=y) = 2/9  P(Temp=h  | Play=n) = 2/5
P(Temp=m | Play=y) = 4/9  P(Temp=m | Play=n) = 2/5
P(Temp=c  | Play=y) = 3/9 P(Temp=c  | Play=n) = 1/5
```

We will also need to calculate the prior probabilities. That is the probability that **play = y** and **play = n**. This is easily done as follows:

```
P(Play=y) = 9/14
P(Play=n) = 5/14
```

Machine Learning Process

Unseen Data

We are in a position to classify previously unseen instances.

Dataset → Machine Learning Algorithm → Model / Hypothesis

We have now built the model

Unseen Data → Model / Hypothesis → Predicts Result

Classify a New Instance

- *Will I play tennis under the following conditions:*
  - ***Outlooks = sunny, Temp = cool, Humidity = high, Windy = true, Play = ?***

Play is y or n. Evaluate probability of each given data.

P(Play = y | Outlook = s, Temp =c, Humidity = h, Wind = t) =
P(Play = y) * P(Outlook = s | Play = y) * P(Temp=c | Play = y) * P(Humidity= h | Play = y) * P(Wind = t | Play = y)
= 9/14 * 2/9 * 3/9 * 3/9 * 3/9 = **0.005291**

P(Play = n | Outlook = s, Temp =c, Humidity = h, Wind = t) =
P(Play = n) * P(Outlook = s | Play = n) * P(Temp=c | Play = n) * P(Humidity= h | Play = n) * P(Wind = t | Play = n)
= 5/14 * 3/5 * 1/5 * 4/5 *3/5 = **0.020571**

Normalise the Results

$$P(c_j)\prod_{x\in X} P(x \mid c)$$

P(Play = y | data) = 0.005291
P(Play = n | data) = 0.020571

(Why do above probabilities not add to 1?)

P(Play = y | data) = (0.005291*100) / (0.005291+ 0.020571) = 20.5%
P(Play = n | data) = (0.020571 *100) / (0.005291+ 0.020571) = 79.5%

Conclusion: more likely NOT to play tennis today.

From the calculations, it is seen that the probability of `Play=yes` is `20.5%`, whereas the probability of `Play=no` is `79.5%`. Selecting the outcome with the higher probability, the classification is that Play=no.

Discussion

Consider the following data instance:

Outlook =overcast, Temp = mild, Humidity = normal, Windy = false: Play = ?

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$$P(c_j)\prod_{x \in X} P(x \mid c)$$

```
P(Outlook=s | Play=y) = 2/9  P(Outlook=s | Play=n) = 3/5
P(Outlook=o | Play=y) = 4/9  P(Outlook=o | Play=n) = 0/5
P(Outlook=r  | Play=y) = 3/9 P(Outlook=r  | Play=n) = 2/5
```

```
P(Wind=t | Play=y) = 3/9   P(Wind=t | Play=n) = 3/5
P(Wind=f | Play=y) = 6/9   P(Wind=f | Play=n) = 2/5
```

```
P(Temp=h | Play=y) = 2/9  P(Temp=h  | Play=n) = 2/5
P(Temp=m | Play=y) = 4/9  P(Temp=m | Play=n) = 2/5
P(Temp=c  | Play=y) = 3/9 P(Temp=c  | Play=n) = 1/5
```

P(Humidity =high| Play = yes) = 3/9
P(Humidity =normal| Play = yes) = 6/9

P(Humidity =high| Play = no) = 3/4
P(Humidity =normal| Play = no) = 1/4

```
P(Play=y) = 9/14
P(Play=n) = 5/14
```

P(Humidity =high| Play = yes) = 3/9
P(Humidity =normal| Play = yes) = 6/9

P(Humidity =high| Play = no) = 4/5
P(Humidity =normal| Play = no) = 1/5

$$P(c_j)\prod_{x \in X} P(x \mid c)$$

Consider the following data instance:

Outlook =<u>overcast</u>, Temp = mild, Humidity = normal, Windy = false: Play = ?

```
P(Outlook=s | Play=y) = 2/9  P(Outlook=s | Play=n) = 3/5
P(Outlook=o | Play=y) = 4/9  P(Outlook=o | Play=n) = 0/5
P(Outlook=r  | Play=y) = 3/9 P(Outlook=r  | Play=n) = 2/5
```

Problem with Using Frequencies for Probability Calculations

- So far we estimated probabilities using the following:
  - $P(X=x1 \mid C=c1) = (N_{x1c1}) / (N_{c1})$
  - N_{x1c1} = counts of cases where attribute X=x1 and class C=c1
  - N_{c1} = count of cases where class C=c1
- What happens when a particular features does not occur for a given class. **In other words when N_{x1c1} = 0?**

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}} P(c_j) \prod_{x \in X} P(x \mid c)$$

- If we multiply all the probabilities together the result of the entire expression becomes 0.

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Discussion

Computer Science Department

Thank You!

Haithem.afli@mtu.ie

Haithem Afli

@AfliHaithem